

Higgs Boson Event Classification Using Machine Learning

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Abstract

This project involves investigating the use of machine learning methods for particle physics inspired event classification. A Random Forest (RF) classifier was trained to separate signal events from background events, based on a multivariate dataset of 10,000 simulated observations, and 20 predictive variables. The process involved data exploration, visualization, model training, model performance evaluation, and feature importance analysis. Ensemble learning methods proved effective for a binary classification problem with 95.9% accuracy for the model.

Introduction

There are large amounts of collision-event data produced in modern particle physics experiments. A key challenge in high energy physics is to detect rare signal events in the presence of many background events.

Machine learning approaches have proven useful in solving classification issues like this. Machine learning models can be used to better identify events of interest by learning complex relationships among multiple measured variables.

This project aimed to develop a full-fledged machine learning pipeline that can separate signal events from background events and assess the performance of the machine learning model using the traditional classification metrics.

Dataset Description

The dataset used in this project consisted of:

- 10,000 simulated events
- 20 numerical predictive features
- Binary target labels

Class definitions:

- Signal Event (1)
- Background Event (0)

The data set was balanced with about equal number of signal and background events, and can be used for a supervised binary classification.

Methodology

The project was carried out in the following process:

Learn how to load and view data. Learn to load and view data. 2. Exploratory data analysis (EDA) 3. Class distribution visualization 4. Feature distribution analysis 5. Train-test splitting 7. Model training using Random Forest. 8. Training the model using the Random Forest model. 7. Model evaluation 8. Confusion matrix analysis 9. Feature importance analysis

The dataset was split into training and testing sets, 80:20.

Training Set:

- 8,000 events

Testing Set:

- 2,000 events
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Exploratory Data Analysis

Initial exploration was performed to get an understanding of the data set and to check the data quality.

Key observations included:

There were no missing values. Features were numeric, and appropriate for machine learning algorithms. The balance of class distribution was maintained. There was some variability between the distributions of the feature events and of the background events.

A few visualizations were created to aid exploratory analysis.

Model Development

Robustness, interpretability, and the relatively strong performance on structured tabular data were key factors in the choice of Random Forest classifier.

Model parameters:

- Number of estimators: 100
- Random state: 42

The training subset was used to train the model and the testing data was used to evaluate it.

Results

Model Performance

The overall classification accuracy of the Random Forest Classifier on testing dataset is **95.9%**.

Performance metrics:

Metric	Value
Accuracy	95.9%
Precision	0.96
Recall	0.96
F1-Score	0.96

The confusion matrix revealed that most events were correctly identified, and there were relatively few false positives and false negatives.

Feature Importance Analysis

Variables affecting most significantly in classification were identified by feature importance analysis. The outcomes revealed the capability of the ensemble machine learning methods to get the meaningful information from the multi-dimensional dataset.

Visualizations Generated

For the analysis the following figures were obtained:

- Class distribution plot
- Feature distribution comparison
- Confusion matrix
- Feature importance ranking

These visualizations helped to gain insight into the structure of these datasets and the behaviour of the models.

Discussion

The model performed well, with high accuracy and a balanced precision and recall.

The results show that machine learning approaches can be used to successfully distinguish signal and background events in structured data. Additionally, feature importance analysis revealed that not every variable was equally important in terms of classification performance and thus the importance of multivariate approaches was emphasized.

This study was based on the simulated data, but the workflow is similar to the ones usually used in particle physics data analysis.

Conclusion

The overall machine learning workflow that was implemented within this project was a successful one for the classification of events.

The accuracy of the classification of signal and background events using a Random Forest classifying was 95.9%. The project provides practical experience in data analysis, machine learning, model evaluation,

scientific visualization, and computational problem solving with Python.

The techniques developed in this work will serve as a basis for further studies of real high-energy physics data.

Future Work

This project has potential for extension to:

- Hyperparameter optimization The method was compared with XGBoost models.
 - Neural network implementation
 - Larger datasets Real Higgs boson event data analysis Advanced Modelling Interpretability techniques
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Limitations

All of the data used in this project was synthetic and is not real data.

In reality, experimental data will have other complexities such as detector effects, statistical uncertainties, and much larger numbers of events. Therefore, the results reported herein are for demonstration purposes only and are not a physics measurement.